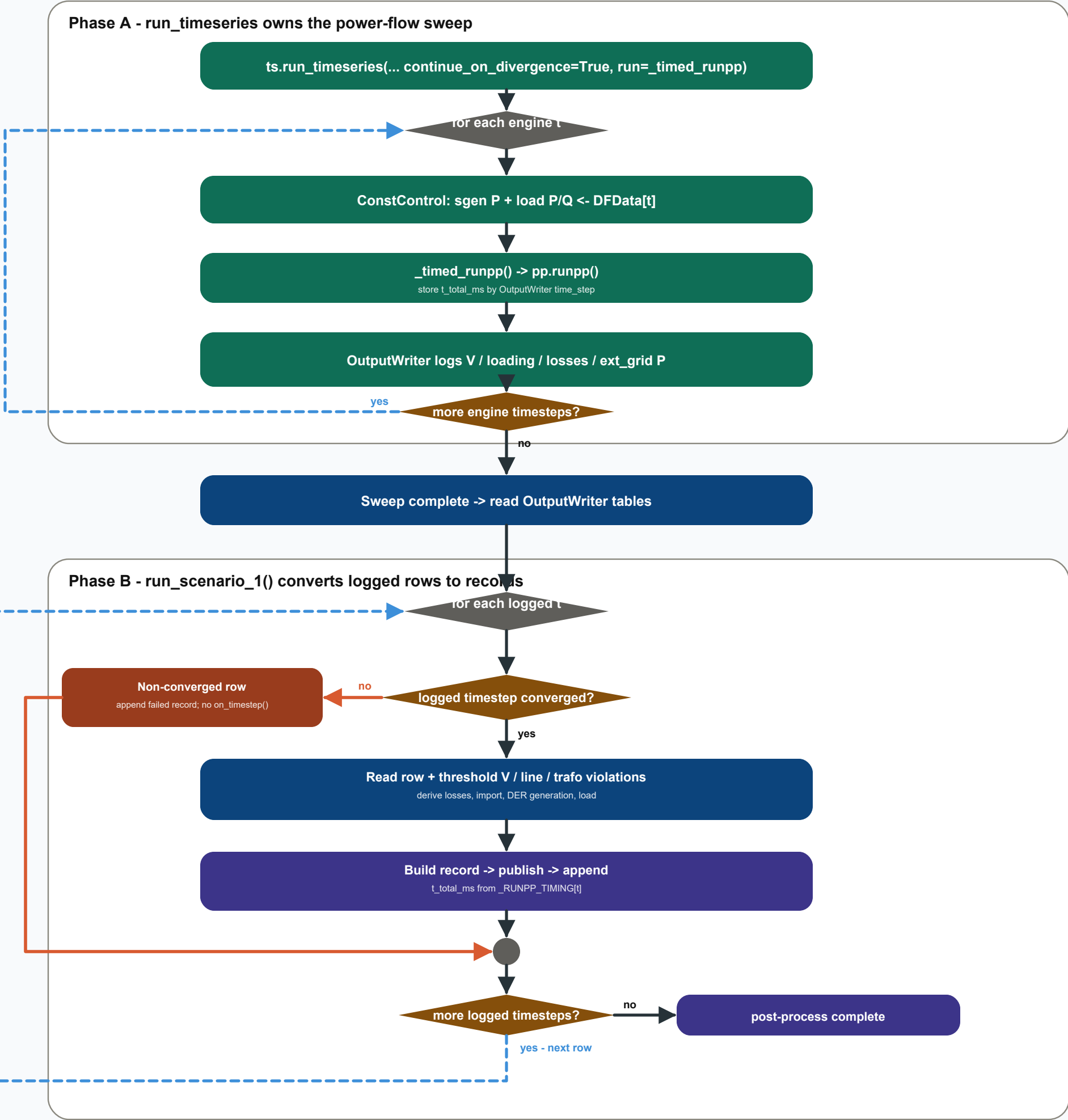




Phase A semantics

- ConstControl is the only per-step state writer in the annual sweep.
- run_timeseries forwards voltage_depend_loads=False to the internal run function.
- continue_on_divergence=True keeps the annual sweep moving after a failed PF timestep.
- _timed_runpp measures only runpp wall time; it does not include post-processing cost.
- OutputWriter is the handoff boundary: violation decisions are not made during this phase.

Phase B record semantics

- Convergence is inferred from the logged vm_pu row: present and not all-NaN.
- Converged rows are re-thresholded against V/load limits during post-processing.
- Losses require both line and trafo loss logs for that timestep; otherwise losses_mw=None.
- Grid import is taken from res_ext_grid.p_mw when the row is available.
- Failed rows get empty result series and are appended immediately without publisher.on_timestep().

Publisher / implementation notes

- No get_resume_records() call exists in Scenario 1, so the runner cannot resume from its checkpoint.
- live_csv_rewrite_fn is accepted but never called; there is no t%96 progress branch.
- on_timestep() occurs only after the annual sweep, while converged rows are being post-processed.
- The runner does not explicitly force sgen q_mvar to zero; it only drives DER p_mw via ConstControl.
- The source comment says _RUNPP_TIMING is cleared per run, but no clear() call appears in run_scenario_1().
- These are implementation-faithful notes, not corrections introduced by the diagram.